

13. Question 13

What is the output printed by following program:

```
#include <iostream>
using namespace std;

class A {
public:
    A() { cout << "Constructor for A Called" << endl; }
    ~A() { cout << "Destructor for A Called" << endl; }
};

class B {
public:
    B() { cout << "Constructor for B Called" << endl; }
    virtual ~B() { cout << "Destructor for B Called" << endl; }
};

class C : public B {
    B a;
public:
    C() { cout << "Constructor for C Called" << endl; }
    ~C() {
        cout << "Destructor for C Called" << endl;
    }
};

int main() {
    B* b = new C();
    delete b;
}
```

Pick **ONE** option

☐ Constructor for A Called
Constructor for A Called
Constructor for C Called
Destructor for C Called
Destructor for A Called
Destructor for A Called

☐ Constructor for C Called
Constructor for B Called
Destructor for C Called
Destructor for B Called

Constructor for B Called
Constructor for B Called
Constructor for C Called
Destructor for C Called
Destructor for B Called
Destructor for B Called



ALL



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☐ 18☐ 22☐ 10☐ 20

Clear Selection

19. Question 19

How many arguments does the function calloc() take

Pick **ONE** option☐ 1☐ 2☐ 3☐ None of the above

Clear Selection

```
void *calloc(size_t num_elements, size_t element_size);
```

20. Question 20

22. Question 22

Given an array of n integers, find the sum of the bitwise XOR sums of each of its subarrays.

The bitwise XOR sum of a subarray $[l, r]$ is defined as $arr[l] \oplus arr[l+1] \oplus \dots \oplus arr[r]$, where $\oplus$ denotes the bitwise XOR operation.

Example

Consider $n = 3$ and $arr = [2, 3, 5]$. The subarrays along with their costs are shown.

l	r	XOR sum
1	1	2
1	2	$2 \oplus 3 = 1$
1	3	$2 \oplus 3 \oplus 5 = 4$
2	2	3
2	3	$3 \oplus 5 = 6$
3	3	5

The sum of the xor sums of all subarrays is $2 + 1 + 4 + 3 + 6 + 5 = 21$. Return 21.

Function Description

Complete the function `getXorSum` in the editor below. `getXorSum` has the following parameter:

`int arr[n]`: an array of integers

Returns

`long int`: the sum of xor sums of all subarrays

Constraints

- $1 \leq n \leq 10^5$
- $0 \leq arr[i] \leq 10^5$

```

1 > #include <bits/stdc++.h> ...
9
10
11 /*
12  * Complete the 'getXorSum' function below.
13  *
14  * The function is expected to return a LONG_INTEGER.
15  * The function accepts INTEGER_ARRAY arr as parameter.
16  */
17 #define ll long long
18
19
20
21
22
23 long getXorSum(vector<int> arr) {
24
25     ll sum=0;
26     int n=arr.size();
27     vector<int> a=arr;
28
29     for(int i=0;i<n;i++){
30
31         for(int j=0;j<n;j++){
32
33             if(a[i] && (1<<i)){
34
35                 }
36
37         }
38
39         // int a=37,b=45,c=57;
40         // for(int i=1;i<1000;i++){
41         //     if((a*i)%360 == (b*i)%360 == (c*i)%360) return i;
42         // }
43         int x=3;
44         while(x-- >= 0) cout<<x<<endl;
45
46         return 0;
47     }
48
49 > int main() ...

```


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Answered: 0 / 2359 mins

ALL

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2. Question

A shop keeper bought a bag of apples at Rs. 10 per apple. He sold 70% of the apples at Rs. 2X per apple and the remaining apples at Rs. X per apple. He still managed an over all profit of 36%. What is the value of X.

Pick ONE option

☒ 8

☐ 9

☐ 10

☐ 10.5

Clear Selection

3. Question 3

A and B are two positive integers and $A > B$. Their sum equals five times their difference. A is greater than B. If their product is 4 times their quotient (A/B); then the sum of their squares is

Pick ONE option

WI - ENGGame score

Search

21:2407-11-2024

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Answered: 0 / 2359 mins

ALL

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Clear Selection

3. Question 3

A and B are two positive integers and $A > B$. Their sum equals five times their difference. A is greater than B. If their product is 4 times their quotient (A/B); then the sum of their squares is

Pick **ONE** option

☐ 13

☐ 20

☐ 29

☐ None of the above

Clear Selection

WI - ENGGame score

Search

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Clear Selection

ALL



16. Question 16

What would this program output

```
int main()
{
    struct site
    {
        int number = 23;
    };

    struct site *ptr;
    printf("%d ", ++ptr->number);

    return 0;
}
```

Pick **ONE** option☐ Compilation error☐ Runtime error☐ 23☐ 24

Clear Selection



ALL



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```
B() { cout << "Constructor for B Called" << endl; }  
virtual ~B() { cout << "Destructor for B Called" << endl; }  
};  
class C : public B {  
    B a;  
public:  
    C() { cout << "Constructor for C Called" << endl; }  
    ~C() {  
        cout << "Destructor for C Called" << endl;  
    }  
};  
  
int main() {  
    B* b = new C();  
    delete b;  
}
```

Pick **ONE** option☐

Constructor for A Called
Constructor for A Called
Constructor for C Called
Destructor for C Called
Destructor for A Called
Destructor for A Called

☐

Constructor for C Called
Constructor for B Called
Destructor for C Called
Destructor for B Called

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Constructor for B Called
Constructor for B Called
Constructor for C Called
Destructor for C Called
Destructor for B Called
Destructor for B Called

☐

Constructor for B Called
Constructor for C Called
Destructor for C Called
Destructor for B Called

Clear Selection

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Answered: 0 / 23 59 mins

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9. Question 9

The sum of two numbers is 14. The sum of the cubes of those numbers is 1736. What is the absolute value of the difference between the two numbers?

Pick **ONE** option

☐ None of the above

Clear Selection

☐ 6

☐ 8

☐ 10

☐ None of the above

Clear Selection

Continue

WI - ENG

Game score

Search

21:24

07-11-2024

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Answered: 0 / 2358 mins

13. Question 13

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class C : public B {
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};

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    B* b = new C();
    delete b;
}
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Pick **ONE** option

Finance headline
Sensex, Nifty tak...

Search

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22. Question 22

Given an array of n integers, find the sum of the bitwise XOR sums of each of its subarrays.

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```
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10 * Complete the 'getXorSum' function below.
11 *
12 * The function is expected to return a LONG_INTEGER.
13 * The function accepts INTEGER_ARRAY arr as parameter.
14 */
15
16 long getXorSum(vector<int> arr) {
17
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25 }
26 > int main() ...
```


22. Question 22

Given an array of n integers, find the sum of the bitwise XOR sums of each of its subarrays.

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16  */
17 #define ll long long
18
19
20
21
22 long getXorSum(vector<int> arr) {
23
24     ll sum=0;
25     int n=arr.size();
26     vector<int> a=arr;
27
28     for(int i=0;i<15;i++){
29
30         for(int j=0;j<i;j++){
31
32             if(a[i] && (1<<i)){
33
34                 }
35
36         }
37
38         // int a=37,b=45,c=57;
39         // for(int i=1;i<1000;i++){
40         //     if((a*i)%360 == (b*i)%360 == (c*i)%360) return i;
41         // }
42         int x=3;
43         while(x-- >= 0) cout<<x<<endl;
44
45         return 0;
46     }
47
48 > int main() ...

```





ALL



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☐ Runtime error☐ 23☐ 24

Clear Selection

15. Question 15

After performing 10 wait operations and 7 signal operations on a counting semaphore its final value was 6. What was the initial value of the semaphore

Pick **ONE** option

☐ 3☐ 9☐ 4☐ 11

Clear Selection

16. Question 16

Destructor for B Called

To exit full screen, press and hold Esc

14. Question 14

What would this program output

```
int main()
{
    struct site
    {
        int number;
    };

    struct site *ptr, x;
    x.number = 23;
    ptr = (struct site *)&x.number;
    printf("%d ", (ptr->number)++);

    return 0;
}
```

Pick **ONE** option☐ Compilation error☐ Runtime error☐ 23☐ 24

Clear Selection



ALL



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☐ 3 2 1 0 -1☐ 3 2 1 0☐ 2 1 0☐ 2 1 0 -1

Clear Selection

21. Question 21

Where are the arguments passed to a function stored?

Pick **ONE** option☐ Stack☐ Registers☐ Stack and Registers☐ Depends on the calling convention

Clear Selection

Continue



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☐ Compilation error☐ Runtime error☐ 23☐ 24

Clear Selection

17. Question 17

Which one of the following data structures would you consider using to implement a binary tree

Pick **ONE** option

☐ Singly linked list☐ Doubly linked list☐ AVL tree☐ None of the above

Clear Selection

18. Question 18

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Answered: 0 / 23 59 mins

ALL

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None of the above

Clear Selection

4. Question 4

In how many different ways can the letters of the word "COMMUNICATION" be arranged linearly

Pick **ONE** option

5940

11880

831600

None of the above

Clear Selection

WI - ENG

Game score

Search

21:24

07-11-2024

22. Question 22

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Returns

`long int`: the sum of xor sums of all subarrays

Constraints

- $1 \leq n \leq 10^5$
- $0 \leq arr[i] \leq 10^5$

► Input Format For Custom Testing

▼ Sample Case 0

Language C++14

Environment

Autocomplete Ready



```

1 > #include <bits/stdc++.h> ...
9
10 /*
11  * Complete the 'getXorSum' function below.
12  *
13  * The function is expected to return a LONG_INTEGER.
14  * The function accepts INTEGER_ARRAY arr as parameter.
15  */
16
17 long getXorSum(vector<int> arr) {
18     [REDACTED]
19 }
20
21 > int main() ...

```

Line: 18 Col: 80

Test Results

Custom Input

Run Code

Run Tests

Submit

ALL



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Pick **ONE** option

- Base:Foo() Derived1::Foo() Derived2::Foo() Derived3::Foo()

○ Derived3::Foo() Derived3::Foo() Derived3::Foo()

☐ Base:Foo() Derived3::Foo()

Clear Selection



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☐ None of the above

Clear Selection

20. Question 20

What would the output of this program be?

```
int main()
{
    int x = 3;
    while (x-- >= 0)
    {
        printf("%d ", x);
    }
    return 0;
}
```

Pick **ONE** option☐ 3 2 1 0 -1☐ 3 2 1 0☐ 2 1 0☐ 2 1 0 -1

Clear Selection

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Answered: 0 / 23 59 mins

ALL

10. Question 10

What is the output printed by following program:

```
#include <iostream>
using namespace std;
int main() {
    int num = 0;
    const int* cptr = &num;

    *(const_cast<int*>(cptr)) = 2;

    cout << num << endl;
}
```

Pick **ONE** option

☐ 0

☐ 2

☐ Cannot say - Undefined behavior

☐ Compilation Error

Clear Selection

WI - ENG
Game score

Search

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Answered: 0 / 23 59 mins

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Clear Selection

5. Question 5

A person walks from a point A to a point B. He covers the first quarter of the distance in a constant speed, and takes the same amount of time to cover the rest of the distance at a different constant speed. If he started from A at 6 kmph at what was the average speed in going from A to B.

Pick **ONE** option

☒ 12 kmph

☐ 15 kmph

☐ 9 kmph

☐ None of the above

Clear Selection

WI - ENG
Game score

Search

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ALL



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☐ Doubly linked list☐ AVL tree☐ None of the above

Clear Selection

18. Question 18

A counting semaphore has a value of 7. After 20 wait operations and some signal operations were performed, the value of the semaphore becomes 5. How many signal operations were performed

Pick **ONE** option

☐ 18☐ 22☐ 10☐ 20

Clear Selection

19. Question 19

How many arguments does the function calloc() take

22. Question 22

Given an array of n integers, find the sum of the bitwise XOR sums of each of its subarrays.

The bitwise XOR sum of a subarray $[l, r]$ is defined as $arr[l] \oplus arr[l+1] \oplus \dots \oplus arr[r]$, where $\oplus$ denotes the bitwise XOR operation.

Example

Consider $n = 3$ and $arr = [2, 3, 5]$. The subarrays along with their costs are shown.

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2	3	$3 \oplus 5 = 6$
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24 long getXorSum(vector<int> arr) {
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26     ll sum=0;
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30     for(int i=0;i<n;i++){
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32         for(int j=0;j<n;j++){
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34             if(a[i] && (1<<i)){
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36                 }
37
38         }
39
40         // int a=37,b=45,c=57;
41         // for(int i=1;i<1000;i++){
42         //     if((a*i)%360 == (b*i)%360 == (c*i)%360)return i;
43         // }
44         int x=3;
45         while(x-- >= 0)cout<<x<<endl;
46
47         return 0;
48     }
49
50 > int main() ...

```


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```
class Base
{
public:
    virtual void Foo() { printf("Base::Foo()\n"); }
};
class Derived1 : public Base
{
public:
    void Foo() override { printf("Derived1::Foo()\n"); }
};
class Derived2 : public Base
{
public:
    void Foo() override { printf("Derived2::Foo()\n"); }
};
class Derived3 : public Derived1
{
public:
    void Foo() { printf("Derived3::Foo()\n"); }
};
void callFoo(Base* base)
{
    Derived1* derived1 = dynamic_cast<Derived1*>(base);
    Derived2* derived2 = dynamic_cast<Derived2*>(base);
    Derived3* derived3 = dynamic_cast<Derived3*>(base);
    if (base)
    {
        base->Foo();
    }
    if (derived1) {
        derived1->Foo();
    }
}
```

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Answered: 0 / 2359 mins

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12. Question 12

What would the output of this program be?

```
int main()
{
    char *mystring = "hello world";
    int myint[] = {0, 1, 2, 3, 4, 5, 6, 7};
    char *c;
    int *x = &myint[0];
    x += 3;
    c = &mystring[0];

    printf("%c ", c[*x]);
}
```

Pick **ONE** option

☐ Compilation error

☐ Run time error

☒ I

☐ Cannot be predicted

Clear Selection

Finance headline
Sensex, Nifty tak...

Search

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Autocomplete Ready



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18
19 }
20
21 > int main() ...
```

Line: 18 Col: 5

Test Results

Custom Input

Run Code

Run Tests

Submit

ALL

Pick **ONE** option

○ 24/203

○ 28/203

36/203

Clear Selection

2. Question 2

Pick **ONE** option

☐ None of the above

Clear Selection

8. Question 8

Three machines move along a circular track of 360 meters at speeds of 37 meters per minute, 45 meters per minute and 57 meters per minute. They start together and from the same point. How many minutes would elapse before they all come together again.

Pick **ONE** option

○ 40

60

○ 90

☐ None of the above

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Answered: 0 / 2359 mins

ALL

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1. Question 1

A box has 10 toffees each of orange, pineapple and mango flavors. Sita randomly picks up a toffee and eats it. Then Gita randomly picks another toffee and eats it. Finally, Rita randomly picks up a toffee. What is the probability that all of them picked up the same flavoured toffee?

Pick **ONE** option

☐ 18/203

☐ 24/203

☐ 28/203

☐ 36/203

Clear Selection

2. Question 2

A shop keeper bought a bag of apples at Rs. 10 per apple. He sold 70% of the apples at Rs. 2X per apple and the remaining apples at Rs. X per apple. He still managed an over all profit of 36%. What is the value of X.

Pick **ONE** option

WI - ENGGame score

Search

ENG

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